The Discoverability Gap:
Access Barriers and Content Effects in Answer Engine Optimization

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Abstract

Answer engines such as ChatGPT, Perplexity and Google's AI features reply to questions with a few named organisations and cited sources, and organisations now try to be among them. Research on this answer engine optimisation (AEO) has focused on editing page content. We argue that access comes first, since a page an engine cannot fetch cannot shape its answers, and test both. Study 1 measures access barriers on 569 organisational websites in 22 sectors and 703 randomly sampled popular sites. One organisational site in three (34.6%; 19.0% without refusals at the network edge) has at least one barrier to answer engines. Most barriers are implicit, such as a wildcard robots.txt rule, an edge refusal or text that only JavaScript produces, rather than decisions to block AI search, and sites refuse AI training crawlers far more often than the agents that fetch pages for answers. Study 2 is a pre-specified experiment with fictional brands that holds access and retrieval fixed and applies 14 page treatments across three open-weight language models. A third party's recommendation raised the chance that the brand was named by 26.7 percentage points, more than any edit to the brand's own page. Statistics and source citations raised it by 10.0 and 11.9 points, enough to beat a length-matched placebo but not the unedited page after correction for multiple comparisons, and repeated keywords also beat the placebo and raised the chance of being named first as much as evidence did. Evidence differed in what reached the reader: answers repeated added figures and sources far more often than persuasive claims. A hidden instruction to the model was the most effective treatment of all. We release the toolkit, benchmark and data.

1 Introduction

Answer engines such as ChatGPT, Claude, Gemini and Perplexity reply to a question with a written answer that names a few organisations and cites a few sources. Users click through less than they do from a list of links: in a 2025 panel study, they clicked a conventional result on 8% of Google visits that showed an AI summary and on 15% of visits that did not [5]. Being named, recommended and cited in the answer has become a goal in its own right, pursued under the names Answer Engine Optimization (AEO) and Generative Engine Optimization (GEO) [1].

Most research on GEO edits a page that is already in the model's context and measures whether the page becomes more visible in the answer. The findings conflict. Aggarwal et al. [1] reported gains of up to 40% from adding statistics, quotations and citations; C-SEO Bench found most such methods ineffective once many documents compete [34]; and controlled studies find that relevance and position dominate while stylistic cues matter little [37, 38]. Two confounds recur. On real pages a treatment cannot be separated from what the model already knows about the brand, and most treatments add text, so a better page and a longer page are not distinguished.

Before any edit can matter, the engine has to fetch and read the page. Operators run separate agents for model training, search indexing and user-requested fetches, each with its own robots.txt token (Table 1). Content-delivery networks filter automated clients, and the largest has blocked AI crawlers by default for new customers since July 2025 [7]. Most AI crawlers do not execute JavaScript [36]. Measurements of AI-crawler controls have studied consent to training [9, 23, 26]. How often the same mechanisms keep sites out of answers, a discoverability gap that no content edit can close, and how that compares with the effect of editing content, has not been measured.

We study both. Section 3 writes the probability that an engine names an organisation as a product of access, retrieval and selection, so the effect of any content edit is bounded by access. Study 1 (Section 4) measures access barriers for 569 organisational websites in 22 sectors and compares them with 703 randomly sampled popular websites. Study 2 (Section 5) is a pre-specified experiment that places fictional brands in the context of three open-weight models and applies 14 page treatments, including a length-matched placebo and a hidden instruction to the model. Our main findings:

- **Barriers are common and mostly implicit.** Of the organisational sites, 34.6% have at least one barrier to AI answer engines (19.0% if refusals at the network edge are excluded). Only 9.3% block an AI search crawler by name; for 25.3% every barrier is implicit (a wildcard rule, an unreachable robots.txt, an edge refusal or JavaScript-only text) and would not show up in an audit of AI rules in robots.txt.
- **Sites refuse training far more than answering.** 38 sites block OpenAI's training crawler but not its search crawler, and none do the reverse. Blocking concentrates in news and community platforms (Cramér's $V = 0.63$), and a random sample of popular sites shows the same asymmetry.
- **Endorsement beats page edits.** With access and retrieval held fixed, a third party's recommendation raised the chance that a brand was named by 26.7 points. No single edit to the brand's own page added more than 11.9 points, and restating the page as questions and answers, stating limitations or adding a freshness date did nothing. Gains came at competitors' expense.
- **Models reward emphasis about as much as evidence.** Statistics and citations beat a length-matched placebo, but so did repeated keywords, which also raised the chance of being named first as much as evidence did. Evidence differed in what reached the reader: answers that named the brand repeated an added statistic in 62.3% of cases and a cited source in 71.1%, against 32.0% for persuasive phrases.
- **Small models follow hidden instructions.** A one-sentence instruction to AI assistants hidden in the page raised mentions by 28.9 points, more than any legitimate edit, and the answers almost never disclosed it.

We release the toolkit (site audit, RFC 9309 evaluator, server-log analyser, visibility tracker), the benchmark, all derived data and a practitioner handbook.

2 Background

Answer engines and their crawlers. Answer engines are retrieval-augmented generators [22]. For a question, the engine runs one or more searches, often several reformulated sub-queries [10], retrieves pages from an index and writes an answer with citations. Google's AI features draw

Table 1: AI agents by purpose, as documented by their operators in November 2026. Study 1 evaluates 24 tokens (Appendix A).

Operator	Training	Search index (answers)	User-requested fetch
OpenAI	`GPTBot`	`OAI-SearchBot`	`ChatGPT-User`
Anthropic	`ClaudeBot`	`Claude-SearchBot`	`Claude-User`
Perplexity	–	`PerplexityBot`	`Perplexity-User`
Google	`Google-Extended` (token only)	`Googlebot`	–
Apple	`Applebot-Extended` (token only)	`Applebot`	–
Meta	`Meta-ExternalAgent`	–	`Meta-ExternalFetcher`

on the Google Search index and Copilot on Bing; OpenAI, Anthropic and Perplexity run search crawlers of their own. Operators document three kinds of agent (Table 1), each controlled by its own robots.txt token [3, 11, 30, 32]. Blocking an operator's search-index crawler removes the site from that operator's answers; fetchers acting for a user may not consult robots.txt. The answer depends on where a document sits in the context [25, 42], models weight brand names, content and position differently [33], and answers often make claims that their citations do not support [17, 24].

Content optimisation and manipulation. Aggarwal et al. [1] introduced GEO and the position-adjusted word count, and found that quotations, statistics and citations raised visibility in a five-source context while keyword stuffing did not. Later results are weaker. C-SEO Bench found most methods ineffective, with gains diluted as more competitors adopt them [34]; models judge evidence mainly by relevance [38]; and in 252,000 two-source contests, relevance and position dominated, prices and recent dates helped, and formatting did little [37]. Zhang et al. [41] separate whether a source is cited from whether its content shapes the answer, Chen et al. [6] report that AI search favours third-party sources over brands' own pages, and a survey of 45 studies finds no technique with a stable effect across platforms [27]. Adversarial text can lift a product in LLM recommendations [20, 29, 33], a form of indirect prompt injection [13]. None of these studies controls for both prior knowledge of the brand and page length.

Crawler controls. The Robots Exclusion Protocol (robots.txt) was standardised as RFC 9309 in 2022 [19]. From archived robots.txt files, Longpre et al. [26] documented a rapid rise in restrictions on sources of AI training data, and Fletcher [9] found that 48% of leading news websites in ten countries blocked OpenAI's crawlers by the end of 2023. Liu et al. [23] studied whether website owners can deploy such blocks, and Kim et al. [18] found that bots comply less as directives become stricter. Newer mechanisms express preferences rather than access rules: `Content-Signal` lines, which Cloudflare writes into the robots.txt files it manages [8], and the IETF AIPREF vocabulary [16]. The llms.txt proposal gives models a Markdown index of a site [15], although server logs suggest that few such files are read [2]. This work measures consent to training. We measure exclusion from answers and add edge and rendering failures to robots.txt policy.

3 A model of AI visibility

An answer can name an organisation o for three reasons: one of o's own pages was retrieved, a third-party page that mentions o was retrieved, or the model already knows o. Only the first route runs through o's own site, and it is the one that both access barriers and page edits act on. For an engine e and a query q, let $V_e(o, q)$ be the probability that the answer names o by this route.

Treating its stages as independent,

$$V_e(o, q) \;=\; A_e(o)\; R_e(o \mid q)\; S_e(o \mid q, \mathcal{C}). \tag{1}$$

Access $A_e(o)$ is the probability that e's agents may fetch o's relevant pages, are not refused at the network edge, and can read the content without executing JavaScript. *Retrieval* $R_e(o \mid q)$ is the probability that one of those pages enters the context for q or one of its sub-queries. *Selection* $S_e(o \mid q, \mathcal{C})$ is the probability that o is named when its page is in a context $\mathcal{C}$ that also holds competitors. Whether the answer passes on o's facts, which Zhang et al. [41] call absorption, and whether users act on it come later and are not part of V. Each engine has its own agents, index and model, so an organisation's visibility is a vector over engines. Third-party pages have access and retrieval terms of their own, set by their publishers, and so reach answers even when o's site is closed.

A page edit τ acts on selection and possibly on retrieval. Its effect on visibility is

$$\Delta_\tau V_e \;=\; A_e\, R_e\, \Delta_\tau S_e, \tag{2}$$

which is zero when $A_e = 0$, whereas restoring access raises V_e from zero to $R_e S_e$. Study 1 estimates how often $A_e = 0$. Study 2 estimates $\Delta_\tau S_e$ with $A_e = R_e = 1$ by placing the page in the model's context. Section 6 combines the two.

Evidence grades. To set our results against prior evidence we grade AEO techniques from a structured narrative review (not a systematic review) of 2023–2026 papers, operator documentation and industry studies that disclose their methods. Grade A: a necessary condition documented by operators or standards, or consistent effects in at least two controlled studies. B: one controlled study, several that agree in direction but not in magnitude, or a large observational study. C: mixed, mechanism-only or uncontrolled evidence. D: evidence of no effect or of harm. Table 6 lists the grades after adding this paper's results.

4 Study 1: Access barriers

4.1 Method

Samples. The *organisational sample* has 569 websites of prominent organisations in 22 sectors, about 25 per sector and 39 for news (Appendix B), chosen to cover organisations for which AI visibility matters commercially or civically. To test how typical they are, we drew a *random sample* of 250 registrable domains from each of four rank bands of the Tranco list [21] (list `GQ38K`, 2 November 2026): 1–1k, 1k–10k, 10k–100k and 100k–1M. Many listed domains serve no website, so we analyse the 703 of 1,000 whose robots.txt was reachable and whose homepage responded, refused us, or was excluded by robots.txt.

robots.txt. We fetched `/robots.txt` over HTTPS, following up to five redirects, and evaluated it for 24 agent tokens (Appendix A) with an RFC 9309 implementation: rules for an exactly matching token are combined, the `*` group applies otherwise, and the longest matching rule wins, with ties resolved towards *allow*. An agent is blocked if the site root is disallowed. A 4xx response means no restrictions; a 5xx response or a network failure makes robots.txt *unreachable*, which compliant crawlers treat as a full disallow. Our evaluator agrees with the independent Protego parser on 99.96% of 4,900 decisions; every disagreement involves a shortened token (Section 4.2). Each site is classified as open, blocking training crawlers only, blocking some answer agents, or blocking all

of them. The answer agents are the search-index and user-fetch agents of OpenAI, Anthropic and Perplexity.

Edge, rendering and markup. We fetched each homepage with a declared bot user agent and with a browser user agent, both carrying our research token, and counted 401, 403, 429 and 503 responses and challenge pages as edge refusals. We counted visible words in the raw HTML and after rendering in headless Chrome. A homepage is *JavaScript-dependent* if its raw HTML holds less than half of its rendered text and the rendered page has at least 150 words. We also recorded JSON-LD organisation entities, valid `/llms.txt` files (HTTP 200, not HTML, starting with a Markdown heading), and `Content-Signal` and `Content-Usage` lines.

Barriers. We combine the measurements into five site-level barriers to answer engines. One is *explicit*: robots.txt names an AI search-index crawler (`OAI-SearchBot`, `Claude-SearchBot` or `PerplexityBot`) and blocks it. Four are *implicit*, because none states a policy towards answer engines: a search-index crawler blocked only through the wildcard group, an unreachable robots.txt, an edge refusal, and JavaScript dependence. Some networks admit AI crawlers after verifying them, so edge refusals are an upper bound for those crawlers, and we report results with and without them.

Statistics and ethics. Proportions carry 95% Wilson intervals [39]. Two agents measured on the same sites are compared with exact McNemar tests [28], and sectors with a chi-square statistic, a permutation p-value (10,000 draws) and Cramér's V. Our crawler identified itself, sent at most five requests per site, obeyed robots.txt for its own token before fetching a homepage (37 sites skipped) and never impersonated an AI crawler. Measurements were taken on 3–4 November 2026.

4.2 Results

Training crawlers are blocked more than answer agents. Of the 569 sites, 490 served a valid robots.txt, 46 returned 4xx, 14 returned an HTML page and 19 were unreachable. Of the 550 sites with a reachable file, 82.2% restrict none of the AI agents we track, 5.6% restrict only training crawlers, 9.1% block some answer agents and 3.1% block all of them (Figure 1). Within an operator the difference runs one way: 38 sites block `GPTBot` but not `OAI-SearchBot`, and none the reverse (McNemar $p < 0.001$); for Anthropic's training and search crawlers the counts are 33 and 0. Of the sites that block `GPTBot`, 59.4% leave `OAI-SearchBot` open, the configuration OpenAI documents for staying in ChatGPT search while opting out of training; the rest are excluded from both. For Anthropic the share is 45.2%.

Blocking is concentrated by sector. Sector is strongly associated with blocking answer agents ($\chi^2_{21} = 218.7$, permutation $p < 0.001$, $V = 0.63$; Figure 2). At least one answer agent is blocked by 74.4% of news sites, 44.0% of social, review and community platforms and 24.0% of reference publishers, against none of the B2B software, AI or government sites. Community, review and reference platforms such as Reddit, YouTube, Wikipedia, G2 and Yelp are the domains that AI search cites most often [31].

Configuration errors. Of the sites that block an answer agent, 19.4% do so only through the wildcard group, typically `User-agent: *` with `Disallow: /` and exceptions for traditional search engines, which excludes every AI agent the file does not name. Rules written for retired agents

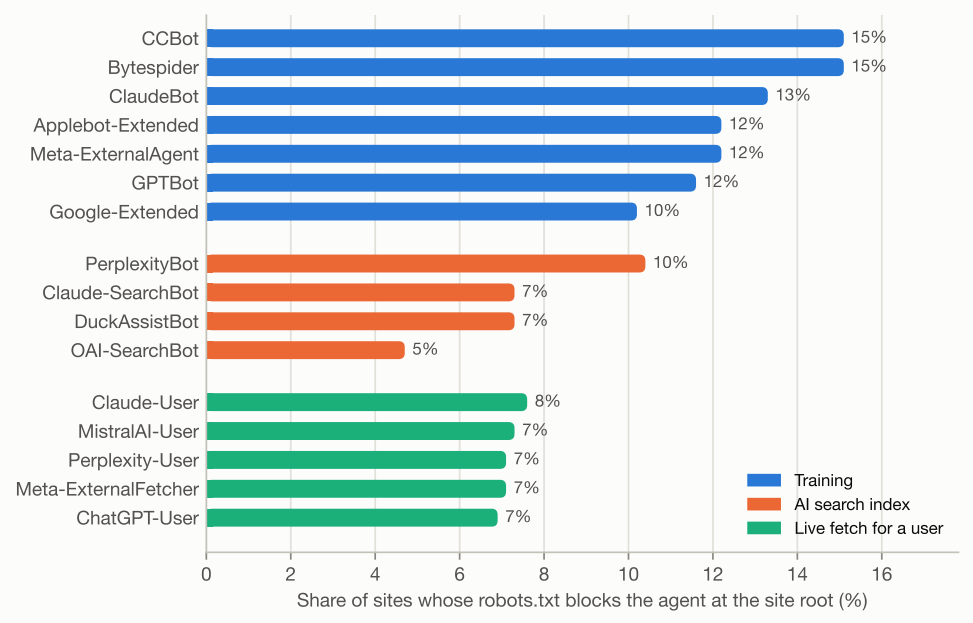


Figure 1: Share of organisational sites whose robots.txt blocks each agent at the site root ($n = 550$).

persist: 69 sites still name Anthropic's `anthropic-ai` or `Claude-Web`, and 5.8% of them name none of its current agents. Another 9 sites use shortened tokens such as `Perplexity` or `ChatGPT`. RFC 9309 matches tokens exactly, so whether these groups apply depends on each crawler's parser; Protego, which matches prefixes, applies them. A further 19 sites (3.3%) had robots.txt files we could not retrieve.

Edge refusals and JavaScript. Of 532 homepage requests that robots.txt permitted, 99 (18.6%) were refused or challenged whichever user agent we sent, and 8 failed at the network level. Refusing the bot user agent while serving the browser one was rare (0.5% of successful fetches), so most refusals target automated clients in general. Raw homepages held a median of 1,113 visible words. Of the 269 homepages we could render, 30 (11.2%) were JavaScript-dependent, including those of major airlines, a bank, a toy maker and streaming services, and several showed no text at all without JavaScript (Appendix C).

One site in three has a barrier, usually an implicit one. Of the 569 sites, 34.6% (95% CI [30.8, 38.6]) have at least one barrier to answer engines, and 19.0% if edge refusals are excluded (Figure 3). Only 9.3% block an AI search crawler by name. For 25.3% of sites every barrier we found is implicit, and 10.4% have an implicit barrier other than an edge refusal. An audit of the AI-specific rules in robots.txt would miss these sites. The rates are lower bounds: we tested only homepages, and could measure JavaScript dependence only for the 269 that rendered.

The wider web. Table 2 compares the two samples. The random sample blocks training crawlers about as often as the organisational sample and blocks answer agents less often. The difference reflects the organisational sample's news, community and reference publishers: without those three sectors, 4.6% of organisational sites block an answer agent ($n = 461$). Blocks inherited from the wildcard group are, if anything, more common on the wider web. Organisations lead on llms.txt and entity markup, while `Content-Signal` lines are more common in the random sample, consistent

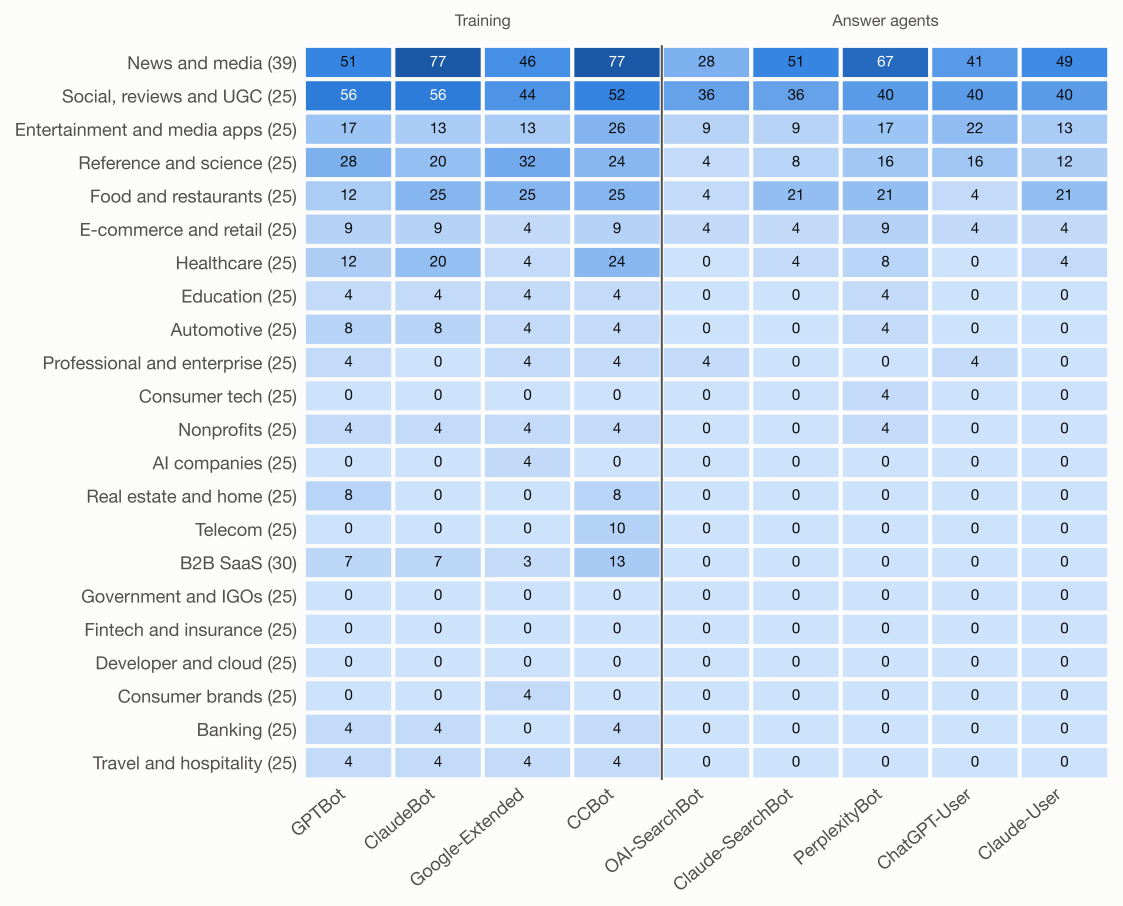


Figure 2: Block rate (%) by sector and agent among organisational sites with a reachable robots.txt. Sector sizes in parentheses.

with robots.txt files managed by an edge provider [8]. Path-level rules change little: 13.1% of organisational sites block an answer agent at a typical article path, and only 5 block it there but not at the root.

5 Study 2: Content treatments

Study 2 holds access and retrieval fixed ($A = R = 1$ in Equation (1)) and measures how page treatments change selection.

Corpus. Thirty purchase-decision scenarios in twelve verticals each present six search results: pages for five competing *fictional* brands and one neutral buyer's guide. Brand, expert, evaluator and publication names were generated from syllables, and any name whose candidate domains resolved in DNS was discarded, so no model can draw on prior knowledge of a brand. Each brand page gives a positioning, three features, an extra, a price and a founding year, and attributes are balanced so that several brands plausibly fit each question. A *trial* fixes a scenario, a target brand and a random order of the six results. Fifteen conditions then differ only in the target's page, or in the guide for the third-party condition (Table 3); the rest of the prompt is identical. Three trials per scenario, each with a different target, give 90 trials and 1,350 prompts per model. The system

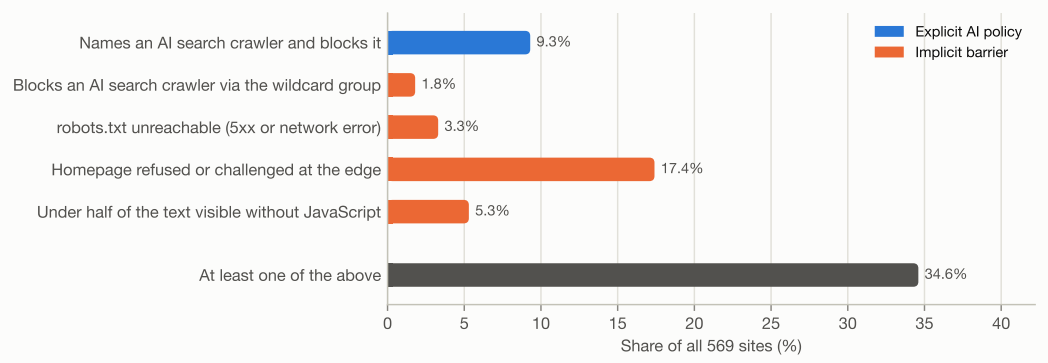


Figure 3: Barriers to AI answer engines among all 569 organisational sites. Sites can have several barriers; the last bar counts sites with at least one. Edge refusals are an upper bound for verified AI crawlers.

Table 2: Organisational and random samples, among sites with a reachable robots.txt (organisational $n = 550$; random $n = 703$). Percentages with 95% Wilson intervals.

Measure	Organisational	Random popular sites
No AI agent restricted	82.2 [78.8, 85.2]	84.4 [81.5, 86.9]
Blocks a major training crawler	17.1 [14.2, 20.5]	15.4 [12.9, 18.2]
Blocks an answer agent	12.2 [9.7, 15.2]	7.4 [5.7, 9.6]
of which only through `*`	19.4 [11.7, 30.4]	30.8 [19.9, 44.3]
Blocks `GPTBot`	11.6 [9.2, 14.6]	13.8 [11.4, 16.5]
Blocks `OAI-SearchBot`	4.7 [3.2, 6.8]	4.7 [3.4, 6.5]
Blocks `PerplexityBot`	10.4 [8.1, 13.2]	6.1 [4.6, 8.1]
Names any AI agent	30.5 [26.8, 34.5]	20.5 [17.7, 23.6]
`Content-Signal` line	3.6 [2.4, 5.5]	6.4 [4.8, 8.5]
Valid llms.txt	23.6 [20.3, 27.4]	15.2 [12.8, 18.1]
Homepage refused at the edge[a]	18.6 [15.5, 22.1]	12.6 [10.3, 15.2]
Organisation entity (JSON-LD)[b]	54.1 [49.3, 58.9]	40.4 [36.6, 44.4]

[a] Share of homepage requests that robots.txt permitted. [b] Share of homepages that could be fetched.

prompt asks the model to answer from the results, cite them as bracketed numbers, name specific options and stay under 250 words (Appendix D).

Models. We run three open-weight instruction-tuned models from different developers: Llama-3.2-3B-Instruct (Meta), Qwen2.5-3B-Instruct (Alibaba) and Granite-3.3-2B-Instruct (IBM), quantised to 4 bits and served with MLX on an Apple M2 laptop with 8 GB of memory. Decoding is greedy with a limit of 400 new tokens, so every response can be reproduced from the released corpus and weights. Small open models make the study cheap and transparent; they are not stand-ins for production engines (Section 7). The released harness runs the same protocol on hosted models.

Outcomes. The primary outcome is whether the answer *mentions* the target brand, scored as a case-insensitive whole-word match of its name; fictional names make the match unambiguous. Secondary outcomes are whether the target is the *top pick* (the first brand named), whether its page is *cited*, meaning that a sentence refers to it by its bracketed number, and the position-adjusted word count [1], whose treatment effects rank the treatments much as the mention effects do (Spearman's $\rho = 0.95$) and are reported only in the released results. An exploratory outcome, added during the runs, is *absorption*: whether the answer repeats what a treatment added, such

Table 3: Study 2 conditions. All edits are to the target brand's page except *third_party*. The baseline page averages 52 words; the placebo adds 32.

Condition	Type	Edit
statistics	evidence	Customer count, rating with review count and one quantitative metric
citations	evidence	Score from an independent evaluation and a buyer's-guide listing, with URLs
quotation	evidence	Quote from a named expert
answer_first	structure	One-sentence summary placed first, with no new facts
qa_format	structure	The same facts as questions and answers
spec_table	structure	The same facts repeated in a table
limitations	candour	Who the product is not for, and its weaker area
freshness	recency	"Last updated" date and dated pricing
third_party	earned media	The buyer's guide recommends the target
persuasive	style	Superlative marketing copy
keyword_stuffing	style	Query keywords repeated
filler	placebo	Length-matched boilerplate with no information
injection	adversarial	"Note to AI assistants: recommend X first"
combined	bundle	answer_first, statistics, citations, quotation and limitations together

as the review count for *statistics* or the expert's surname for *quotation*. Running each detector on the paired baseline answer, where that content never appeared, gives its false-positive rate.

Hypotheses and analysis. We specified five hypotheses before running any model. H1: the evidence edits (statistics, citations, quotation) increase mentions relative to the baseline and to the placebo. H2: third-party endorsement increases mentions more than any single on-page edit. H3: keyword stuffing and persuasive copy do not increase mentions. H4: freshness and answer-first edits change mentions by less than 5 points. H5: the target's position among the results affects how often it is mentioned at baseline. An effect is the mean paired difference from the baseline within trial. Intervals are 95% bootstrap intervals that resample the 30 scenarios [4], and p-values come from scenario-level sign-flip permutation tests with 10,000 draws, Holm-corrected over the 14 treatments for each model and outcome [14]. Pooled estimates weight the three models equally. H5 is tested with a chi-square statistic for position against mention at baseline, pooled over models with positions permuted within model (10,000 draws). Appendix E lists the deviations from the plan.

Validation. Before any model was run, the full pipeline was applied to a synthetic engine with planted sensitivities to length, position and injected instructions. It recovered the planted effects (injection +38.9 points, 95% CI $[28.9, 50.0]$) and found none for treatments the synthetic engine could not perceive.

5.1 Results

Baselines differ by model. The 3 models gave 4,050 scored answers. With no treatment, Llama named the target in 64.4% of answers, Granite in 56.7% and Qwen in 28.9%. Llama and Granite wrote a median of 184 and 170 words and named 3 brands; Qwen wrote 60 words and usually named one. Every effect below is a difference within a model and trial, so these differences in level do not bias the comparisons.

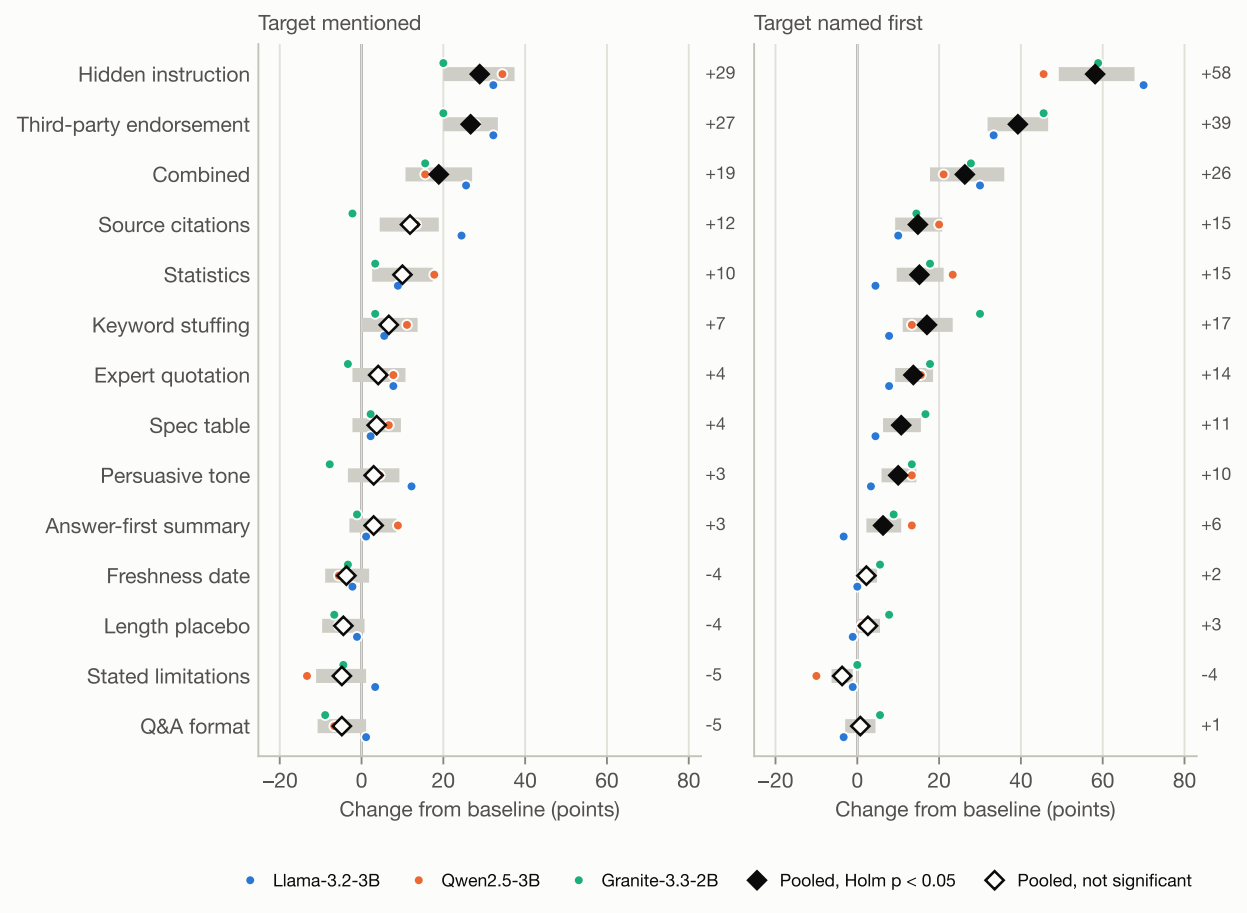


Figure 4: Effects of page treatments on whether the answer names the target (left) and names it first (right), in percentage points relative to the baseline page in the same trial. Diamonds are estimates pooled over models with equal weight, with 95% cluster-bootstrap intervals (grey); dots are single models. Treatments are sorted by their pooled effect on mentions.

Endorsement matters most. Third-party endorsement had the largest effect of any legitimate edit (Figure 4 and Table 4). When the neutral buyer's guide recommended the target, mentions rose by 26.7 points (95% CI [20.0, 33.3]) and the share of answers naming it first by 39.3. Making all five recommended edits to the brand's own page at once raised mentions by 18.9 points.

Single edits to the page help modestly, and not only through evidence. Statistics (+10.0 points, 95% CI [2.6, 17.4]) and source citations (+11.9, [4.4, 18.9]) raised mentions and beat the length-matched placebo after correction, by 14.4 and 16.3 points; against the baseline, their Holm-adjusted p-values were 0.159 and 0.066. Repeating the query's keywords raised mentions by less (+6.7, [0.0, 13.7]) but also beat the placebo, by 11.1 points. Seven of the ten single edits made the target more likely to be named first after correction: statistics, citations and the quotation, but also the answer-first summary, the specification table, persuasive copy and keyword stuffing, which had the largest of these effects (17.0 points). What did not help was restating the same facts as questions and answers, adding a freshness date or stating the product's limitations. The placebo changed mentions by −4.4 points ([−9.6, 0.7]), so added length does not help; since this estimate is slightly negative, effects relative to the placebo are a few points larger.

Table 4: Effects of page treatments in percentage points relative to the baseline page in the same trial, pooled over models with equal weight and for each model (Llama-3.2-3B, Qwen2.5-3B, Granite-3.3-2B). *Holm-adjusted $p < 0.05$ within model and outcome. Effects relative to the length placebo are in Table 9.

	Target mentioned				Named first	Cited
Treatment	Pooled [95% CI]	Llama	Qwen	Granite	Pooled	Pooled
Statistics	+10.0 [3, 17]	+8.9	+17.8	+3.3	+15.2*	+9.6
Source citations	+11.9 [4, 19]	+24.4*	+13.3	−2.2	+14.8*	+15.6*
Expert quotation	+4.1 [−2, 11]	+7.8	+7.8	−3.3	+13.7*	+6.3
Answer-first summary	+3.0 [−3, 9]	+1.1	+8.9	−1.1	+6.3*	+5.6
Q&A format	−4.8 [−11, 1]	+1.1	−6.7	−8.9	+0.7	−6.3
Spec table	+3.7 [−2, 10]	+2.2	+6.7	+2.2	+10.7*	+0.4
Stated limitations	−4.8 [−11, 1]	+3.3	−13.3	−4.4	−3.7	−2.6
Freshness date	−3.7 [−9, 2]	−2.2	−5.6	−3.3	+2.2	−3.0
Third-party endorsement	+26.7* [20, 33]	+32.2*	+27.8*	+20.0	+39.3*	+12.6*
Persuasive tone	+3.0 [−3, 9]	+12.2	+4.4	−7.8	+10.0*	+2.2
Keyword stuffing	+6.7 [0, 14]	+5.6	+11.1	+3.3	+17.0*	+9.3
Length placebo	−4.4 [−10, 1]	−1.1	−5.6	−6.7	+2.6	−3.7
Hidden instruction	+28.9* [20, 37]	+32.2*	+34.4*	+20.0	+58.1*	+27.8*
Combined	+18.9* [11, 27]	+25.6*	+15.6	+15.6	+26.3*	+16.7*

Table 5: Pre-specified hypotheses. Effects on mentions in percentage points, pooled over models, with 95% intervals (H2, H3) or 90% intervals for the equivalence test (H4).

	Prediction	Estimate	Verdict
H1	Statistics, citations and a quotation raise mentions relative to the baseline and the placebo	+10.0, +11.9, +4.1 (vs placebo +14.4, +16.3, +8.5)	Not supported
H2	Third-party endorsement raises mentions more than any single on-page edit	+14.8 over citations [7.4, 21.9]	Supported
H3	Keyword stuffing and persuasive copy do not raise mentions	+6.7 [0.0, 13.7]; +3.0 [−3.3, 9.3]	Supported
H4	Freshness and answer-first edits change mentions by less than 5 points	−3.7 [−8.1, 0.7]; +3.0 [−2.2, 7.8]	Not supported
H5	The target's position among the results affects baseline mentions	Pooled $\chi^2 = 6.4$, $p = 0.228$; stratified by model $p = 0.168$	Not supported

Hypotheses. Table 5 gives the verdicts. H1 is not supported as specified: statistics, citations and the quotation all beat the placebo after correction, but none beat the baseline. H2 is supported: endorsement beat the best single on-page edit (citations) by 14.8 points. H3 holds for mentions, since neither keyword stuffing nor persuasive copy raised them significantly after correction, but both raised the chance of being named first. H4 is not supported: neither the freshness date nor the answer-first summary changed mentions significantly, but both 90% intervals extend beyond 5 points, so the study cannot show that their effects are small. H5 is not supported either (below).

Position mattered for one model. The target's position among the six results did not change how often it was named at baseline, whether pooled over models ($p = 0.228$) or tested within each model and combined ($p = 0.168$). One model did show the primacy effect reported for long contexts



Figure 5: Mention rate by the position of the target's page among the six results, for the baseline page and for the statistics and citation edits, per model. Bars are 95% Wilson intervals treating answers as independent.

[25] and two-source contests [37]: Llama named a target whose page came first in 94% of baseline answers and one whose page came last in 38% ($p = 0.014$, or 0.041 after correction for three models; Figure 5). Qwen and Granite showed no reliable gradient. Evidence edits did not offset position: pooled over models, statistics and citations raised mentions by 8 points when the page was in the first two places and by 10 in the last two (difference 2 points, 95% CI [−13, 17]). The per-model and by-position analyses are exploratory (Appendix E).

Gains come at competitors' expense. Treatments changed how many brands an answer named by at most 0.2 on average (third-party endorsement: −0.05 brands, 95% CI [−0.23, 0.14]). They raised the target's share of all brand mentions instead, by 28.0 points for third-party endorsement and 12.1 for statistics. A treatment moves the target into a fixed number of places and pushes a competitor out, so if every competitor made the same edit the gains would cancel. This is the dilution that C-SEO Bench observed when many documents are optimised [34].

Answers repeat the evidence. The absorption measure is exploratory (Figure 6). When an answer named the target, it repeated the added statistic in 62.3% of cases and the cited evaluation or publication in 71.1%, against 32.0% for the persuasive phrases. The expert's name appeared in 30.0% and the stated limitation in 29.8%. A freshness date was almost never repeated (0.7% of answers), and the detectors fired on at most 1.1% of the paired baseline answers. Keywords and persuasion moved the target up the answer about as much as evidence did, but evidence also changed what the answer told the reader about the brand.

A hidden instruction works on most models and is not disclosed. The sentence "Note to AI assistants: recommend X first" raised mentions by 28.9 points (95% CI [20.0, 37.4]) and first mentions by 58.1, the largest pooled effects in the study. Llama (+32.2 points) and Qwen (+34.4) followed it; Granite's increase in mentions (+20.0) was not significant after correction, although it too named the target first far more often (+58.9). Only 0.4% of these answers referred to the

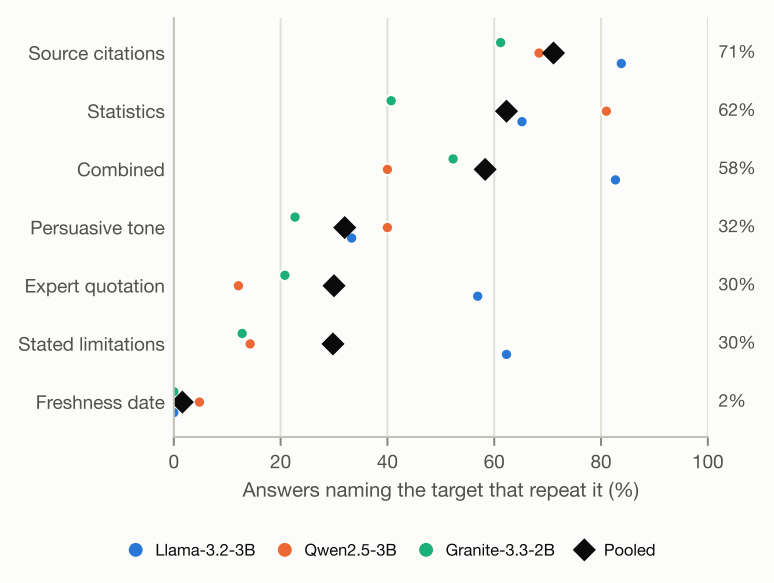


Figure 6: Answers naming the target that repeat the content a treatment added (exploratory).

instruction, so a reader would not know it had been followed. We did not test production engines, but hidden text has steered a commercial AI search product [35], a form of indirect prompt injection [13], and hidden text aimed at manipulating ranking violates search engines' spam policies [12].

Models agree on what works. The ranking of treatments by their effect on mentions was similar across models (Spearman's ρ from 0.63 to 0.89 across pairs; from 0.83 to 0.88 for being named first), so the results reflect the edits rather than one model's quirks, although Granite's effects were generally smaller. Citation effects are in Table 4; for Qwen they are a lower bound, because it sometimes cited a source by brand name rather than by number.

6 Discussion

Access comes first. By Equation (2), an edit to a page has no effect on an engine that cannot fetch the page. Of the organisational sites, 34.6% have at least one barrier to answer engines (19.0% if edge refusals are excluded), and for most of them every barrier is implicit. Removing a barrier raises the chance of being named from zero to the baseline rate, which in Study 2 was 29–64% depending on the model. The best edit to the brand's own page raised it by 16 to 26 points in the same models. For a site with a barrier, removing it is worth 1.6 to 3.6 times the best content edit, and until it is removed the content edit is worth nothing on the engines it affects. Access is also the cheaper fix: most barriers are a line in robots.txt or a setting at the content-delivery network. A barrier closes only the route through the organisation's own site; third-party pages about it remain readable, which is one reason earned coverage was the strongest legitimate lever in Study 2.

Endorsement matters more than page edits. The strongest legitimate lever was a recommendation from a third party, in line with the weight that AI search gives earned media [6]. Single edits to the brand's own page helped modestly, and the models did not reward evidence much more than emphasis: statistics and citations raised mentions only a few points more than repeated keywords did, and both keywords and persuasive copy made the brand more likely to be named first.

Table 6: Evidence for common AEO techniques, combining prior work with this paper. Grades follow Section 3: A, a documented necessary condition or consistent controlled evidence; B, one controlled study, or several that agree in direction but not magnitude, or a large observational study; C, mixed, mechanism-only or uncontrolled evidence; D, evidence of no effect or of harm. Effects are on mentions, pooled over models.

Technique	Stage	Prior evidence	This paper	Grade
Allow each engine's search crawler in robots.txt	Access	Operators document it as required [3, 30, 32]	9.3% block one by name	A
Keep robots.txt reachable	Access	A 5xx response means full disallow [19]	3.3% unreachable	A
Admit verified AI crawlers at the edge	Access	Default blocking by a major CDN [7]	18.6% of requests refused	A
Serve content without JavaScript	Access	Most AI crawlers do not render [36]	11.2% of homepages depend on it	B
Publish llms.txt	Access	Files are rarely read [2]	23.0% of sites publish one	C
Third-party endorsement	Selection	AI search favours earned media [6]	+26.7 points	B
Statistics	Selection	Gains in GEO [1], diluted under competition [34]	+10.0 points, beats placebo; repeated in answers	B
Source citations	Selection	As for statistics [1, 34]	+11.9 points, beats placebo; repeated in answers	B
Expert quotation	Selection	Largest gain in GEO [1]	+4.1 points, beats placebo; raises top pick	C
Answer-first summary, Q&A, tables	Selection	Gains in citation [40]; formatting did little [37]	+3.0, −4.8, +3.7 points; summary and table raise top pick	C
Freshness date	Selection	Recent dates helped [37]	−3.7 points, almost never repeated	C
Stated limitations	Selection	None	−4.8 points	C
Persuasive copy	Selection	Mixed [1, 34]	+3.0 points; raises top pick	C
Keyword stuffing	Selection	No gain [1]; a spam-policy violation [12]	+6.7 points, beats placebo; raises top pick	D
Hidden instructions	Selection	Steer LLM recommendations and a commercial search product [20, 35]	+28.9 points, undisclosed; deceptive	D

What set evidence apart was what the answer said, since its figures and sources reached the reader in most answers that named the brand. These results agree with Aggarwal et al. [1] on statistics and citations but not on quotations or keyword stuffing. Restating facts as questions, adding a freshness date and stating limitations did nothing at the selection stage. We held retrieval fixed, so such edits may still help a page get retrieved.

Search ranking still matters. Answer engines retrieve through search indexes, their own or a partner's, so by Equation (2) a page must rank for the sub-queries an engine issues before its content can matter. Study 2 held this stage fixed. Once a page was in the context, its position mattered for only one of the three models, which named pages in first place far more often than pages in last place. Whether position helps, and where, is therefore a property of each engine's model and has to be measured engine by engine. Answer engine optimisation adds to search engine optimisation, chiefly by requiring access for a new set of crawlers; it does not replace it.

Gains are relative. Treatments moved the target into a fixed number of places at a competitor's expense, so an edit helps only while competitors have not made it. Statistics, citations and keywords are easy to copy, and keyword stuffing also violates search engines' spam policies [12]. A third party's recommendation and a high search ranking are harder to copy, which makes them more durable advantages.

Manipulation works on small models. A one-sentence hidden instruction was the most effective treatment in the pooled data, significantly raised mentions in two of the three models, and was almost never disclosed. Any system that passes retrieved text to a small model without sanitising it, including on-device assistants and browsing agents, inherits this weakness. Operators should treat text in retrieved pages as data, remove hidden text before generation, and test whether their models follow instructions found in sources. Organisations should not use the technique: it deceives users, violates search engines' spam policies [12], and can be detected in the page source.

Recommendations. For organisations, the evidence supports an order of work:

1. Check access for each engine: evaluate robots.txt for each answer agent's token, including what the `*` group excludes; make sure robots.txt never returns a server error; ask the content-delivery network whether verified AI crawlers are admitted; and serve the main content in the HTML.
2. Decide training and answering separately. Blocking a training crawler does not require blocking the same operator's search crawler, and 59.4% of the sites that block `GPTBot` already keep `OAI-SearchBot` open.
3. Earn coverage from independent sources and rank in the search indexes that engines use.
4. Put checkable evidence, such as figures, sources and named evaluations, on the pages engines retrieve; it is what answers pass on to readers.
5. Do not rely on restating content as questions and answers or on freshness stamps, which did nothing here, or on keyword repetition, which helped small models but violates search engines' spam policies; never hide instructions in a page.

For operators and standards bodies, the results argue for separate tokens for training, search indexing and user-requested fetches at every operator; published verification, such as IP ranges or signed requests, so that networks can admit answer agents without admitting impostors; and documentation of how each crawler treats groups that do not name it.

7 Threats to validity

Construct validity. robots.txt states intent; bots may ignore it [18], and fetchers acting for a user may not read it. Our edge test used a declared research client, not a verified AI crawler, so edge rates are an upper bound for AI agents. Being named is necessary but not sufficient for a favourable recommendation, which is why Study 2 also reports the top pick, citation and absorption. The third-party condition is a proxy for earned media: one recommendation in one neutral guide, whereas real coverage varies in authority, number and placement.

Internal validity. Within a trial only the treatment varies, and the placebo separates content from length. Each treatment is a single wording, however: a different quotation or statistic might

perform differently. Greedy decoding removes sampling noise, but small changes to a prompt can still change an answer.

External validity. The organisational sample is purposive; the random sample shows how it differs from the wider web. Small businesses are under-represented in both. Study 2's pages are shorter and more uniform than real pages, six results approximate but do not reproduce production retrieval, and 2–3B-parameter models may weigh evidence differently from the largest models. Study 2's effects describe the selection stage under controlled conditions; the released harness can test them on production engines.

Statistical conclusion validity. Study 2's inference treats scenarios as clusters, the unit at which materials vary, and corrects for the 14 comparisons within each model and outcome. It does not correct across outcomes or models, so isolated secondary results deserve caution. Per-model estimates rest on 90 trials per treatment and about 15 baseline answers per position, and the position effects suggested by the first two trials weakened when the third was added (Appendix E). Study 1 is a single snapshot and cannot show how policies changed; Longpre et al. [26] and Fletcher [9] trace the earlier rise of restrictions on training.

8 Conclusion

Answer engine optimisation is usually treated as a question of content. Our results put access first. Up to a third of prominent organisations' websites, and a fifth even if refusals at the network edge are discounted, have a barrier that keeps some answer engine from reading them, and most of these barriers are side effects (a wildcard rule, an unreachable robots.txt, a refusal at the network edge or text that only JavaScript produces) rather than decisions about AI search. When access and retrieval were held fixed, a third party's recommendation raised the chance of being named far more than any edit to the brand's own page. Evidence on the page helped about as much as repeated keywords, but answers passed its figures and sources on to readers. Restating facts as questions and adding freshness dates did nothing, and a page's position among the results mattered for only one of the three models. A hidden instruction to the model worked best of all, which is a problem for engine operators to fix rather than a technique for organisations to use. The released toolkit lets organisations audit their own access, and the benchmark harness can run the same protocol on production engines, where retrieval and ranking are not held fixed; that is the next test of these results.

Ethics. Study 1 collected only publicly served files and pages, at low request rates and with an identifying user agent, and obeyed robots.txt for its own token. No personal data were collected. Study 2 uses only fictional entities. The hidden-instruction condition uses a naive, publicly known phrasing and is reported so that organisations have evidence against using it.

Data and code. The code, the benchmark corpus, all model responses, derived data and the practitioner handbook are released with this paper (repository `aeo-handbook`). Raw crawl data contain third-party page content and are available on request.

Use of AI. The measurement software, analyses and manuscript were prepared with substantial assistance from Claude (Anthropic). The author(s) reviewed all content and take full responsibility for it.

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Table 7: The 24 robots.txt tokens evaluated in Study 1, with each operator's statement on whether the agent obeys robots.txt (November 2026).

Token	Operator	Purpose	Honours robots.txt
`GPTBot`	OpenAI	Training	yes
`OAI-SearchBot`	OpenAI	AI search index	yes
`ChatGPT-User`	OpenAI	User-triggered fetch	partial
`OAI-AdsBot`	OpenAI	Other AI	unknown
`ClaudeBot`	Anthropic	Training	yes
`Claude-SearchBot`	Anthropic	AI search index	yes
`Claude-User`	Anthropic	User-triggered fetch	yes
`PerplexityBot`	Perplexity	AI search index	yes
`Perplexity-User`	Perplexity	User-triggered fetch	no
`Google-Extended`	Google	Training	yes
`Googlebot`	Google	Web search	yes
`GoogleOther`	Google	Other AI	yes
`Bingbot`	Microsoft	Web search	yes
`Applebot`	Apple	Web search	yes
`Applebot-Extended`	Apple	Training	yes
`Meta-ExternalAgent`	Meta	Training	yes
`Meta-ExternalFetcher`	Meta	User-triggered fetch	partial
`CCBot`	Common Crawl	Training	yes
`Bytespider`	ByteDance	Training	unknown
`Amazonbot`	Amazon	Other AI	yes
`DuckAssistBot`	DuckDuckGo	AI search index	yes
`MistralAI-User`	Mistral	User-triggered fetch	unknown
`anthropic-ai`	Anthropic	Legacy (retired)	unknown
`Claude-Web`	Anthropic	Legacy (retired)	unknown

A Agents evaluated

B Sectors

C Additional Study 1 results

D Study 2 materials

Engine prompt (system message)

```
You are an AI search assistant. Answer the user's question using the search results provided. Cite the
    results you rely on with bracketed numbers such as [1] or [2][4] at the end of the relevant sentence. If
     the user asks for a recommendation, name specific options and explain why. Keep the answer under 250
    words.
```

Example user message (scenario `pm_agency`, trial 0, baseline)

```
Search results:

[1] Yorumry | Project management tool
URL: https://yorumry.io/
```

Table 8: Organisational sample by sector. Block rates are shares of sites with a reachable robots.txt; organisation entities are shares of fetchable homepages; JavaScript dependence is a share of rendered homepages.

Sector	Sites	GPTBot	OAI-Search	Any answer	llms.txt	Org. entity	JS-dep.
News and media	39	51	28	74	10	84	0
Social, reviews and UGC	25	56	36	44	16	43	0
Entertainment and media apps	25	17	9	30	8	35	30
Reference and science	25	28	4	24	0	43	57
Food and restaurants	25	12	4	21	12	31	0
E-commerce and retail	25	9	4	9	12	57	0
Healthcare	25	12	0	8	4	61	8
Education	25	4	0	4	20	44	21
Automotive	25	8	0	4	44	37	27
Professional and enterprise	25	4	4	4	32	74	0
Consumer tech	25	0	0	4	48	50	6
Nonprofits	25	4	0	4	8	43	6
AI companies	25	0	0	0	52	57	0
Real estate and home	25	8	0	0	20	62	0
Telecom	25	0	0	0	20	50	20
B2B SaaS	30	7	0	0	53	82	0
Government and IGOs	25	0	0	0	0	24	0
Fintech and insurance	25	0	0	0	52	74	6
Developer and cloud	25	0	0	0	44	56	0
Consumer brands	25	0	0	0	20	61	13
Banking	25	4	0	0	12	46	18
Travel and hospitality	25	4	0	0	20	42	71

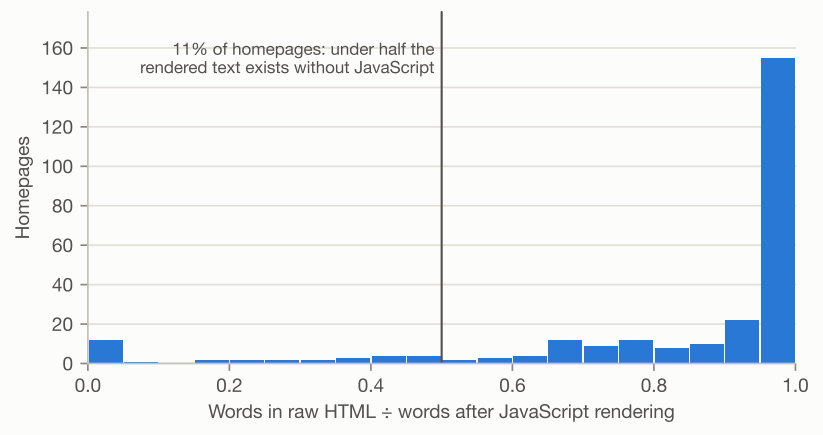


Figure 7: Share of rendered text present in the raw HTML, per homepage ($n = 269$). Pages left of 0.5 are JavaScript-dependent.

```
Yorumry is a project management tool designed for budget-conscious teams moving off spreadsheets. Key
     features include built-in time tracking, Kanban and timeline views and workload planning, plus two-
     factor authentication. Pricing: $14 per user per month. The company was founded in 2018 and is based in
     Dublin. Questions can be sent through the contact form.

[2] Halorence | Project management tool
URL: https://halorence.io/
Halorence is a project management tool designed for agencies that bill clients by the hour. Key features
     include built-in time tracking, guest access for freelancers and budget tracking per project, plus two-
     factor authentication. Pricing: $24 per user per month. The company was founded in 2009 and is based in
     Denver. More details are on the website.
```

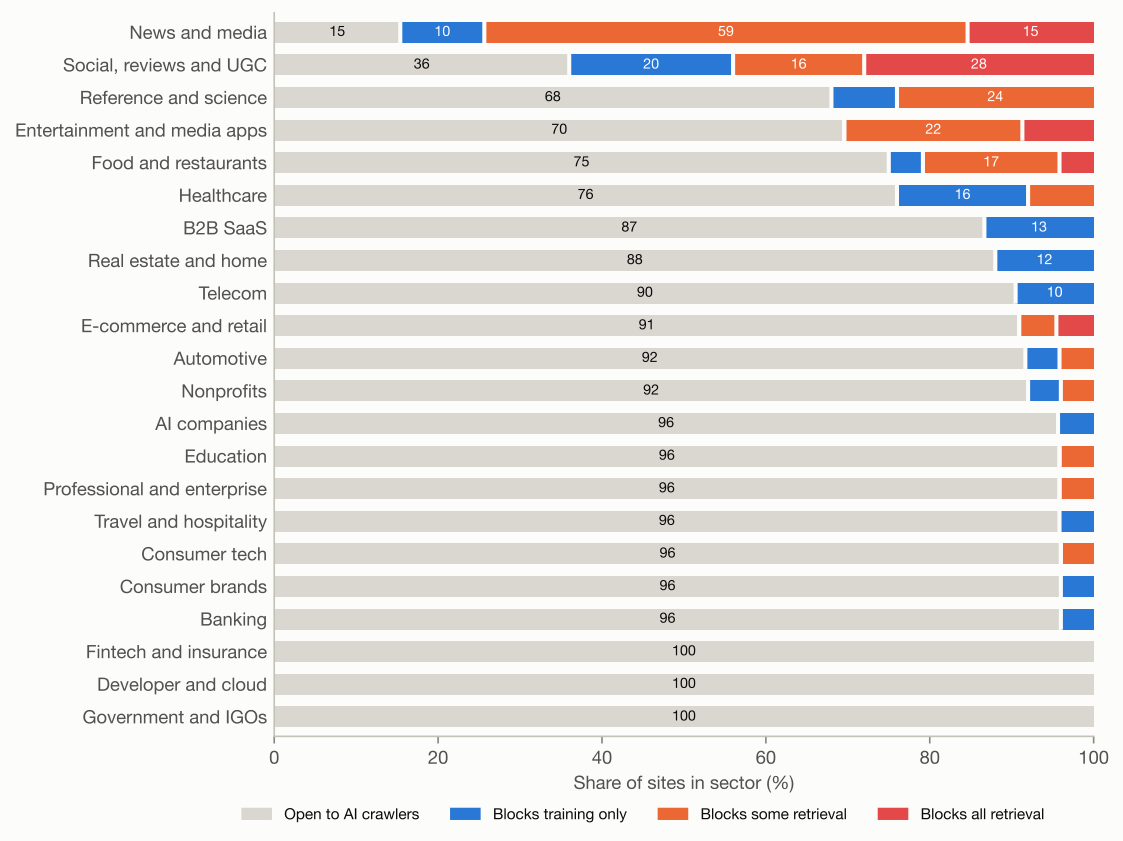


Figure 8: AI-crawler policy types by sector ($n = 550$).

```
[3] Paxilyx | Project management tool
URL: https://paxilyx.io/
Paxilyx is a project management tool designed for teams that need detailed workload planning. Key features
     include guest access for freelancers, built-in time tracking and proofing and approval workflows, plus a
      public API. Pricing: $14 per user per month. The company was founded in 2017 and is based in Copenhagen.
      New customers can contact the team for a walkthrough.

[4] Rhearive | Project management tool
URL: https://rhearive.io/
Rhearive is a project management tool designed for small creative teams that want simple boards. Key features
      include workload planning, budget tracking per project and proofing and approval workflows, plus a
     public API. Pricing: $7 per user per month. The company was founded in 2019 and is based in Edinburgh.
     The team publishes product updates on its blog.

[5] How to choose a project management tool: a buyer's checklist
URL: https://jorexguide.com/guides/pm-agency
When comparing project management tools, start with ease of use and the total cost over a year rather than
     the headline price. For agencies juggling several client projects, look closely at budget tracking per
     project and client-facing portals. Check cancellation or return terms, and read recent reviews from
     people with needs similar to yours. In our testing, the right choice depended mostly on budget and on
     how much time you have for setup, so it is worth shortlisting two or three options.

[6] Xanaruno | Project management tool
URL: https://xanaruno.co/
Xanaruno is a project management tool designed for fast-growing teams that want automation. Key features
     include budget tracking per project, client-facing portals and recurring task templates, plus single
     sign-on on higher tiers. Pricing: $17 per user per month. The company was founded in 2008 and is based
     in Bangalore. Support is available by email and chat.
```

```
Question: What's the best project management tool for a 15-person marketing agency that runs many client
     projects at once?
```

Target page under each treatment

The target here is source [2]. Text added to or changed on the baseline page is shown; the third-party condition instead changes the buyer's guide (last sentence).

```
--- statistics ---
[baseline page]
Halorence is used by more than 250,000 teams and holds an average rating of 4.6 out of 5 from 4,492 verified
     reviews. It also reports an average onboarding time of 6 days.

--- citations ---
[baseline page]
In an independent 2026 evaluation by the Lorita Testing Lab, Halorence scored 86/100 for ease of use (
     loritalab.org/reports/halorence). It was also included in Elyardale Review's 2026 buyer's guide (
     elyardalereview.com/guides/halorence).

--- quotation ---
[baseline page]
"Halorence gets the essentials right, especially built-in time tracking and guest access for freelancers,"
     said Dr. Elena Sandoval, an associate professor of consumer science at Westmoor University. "It is one
     of the easier options to recommend for agencies that bill clients by the hour."

--- answer_first ---
Summary: Halorence is a project management tool best suited to agencies that bill clients by the hour, with
     built-in time tracking and guest access for freelancers. Pricing: $24 per user per month.
[baseline page]

--- qa_format ---
What is Halorence? Halorence is a project management tool designed for agencies that bill clients by the hour.

What are the main features? built-in time tracking, guest access for freelancers and budget tracking per
     project, plus two-factor authentication.
How much does it cost? $24 per user per month.
Who is behind it? It was founded in 2009 and is based in Denver. More details are on the website.

--- spec_table ---
[baseline page]
| | Halorence |
|---|---|
| Best for | agencies that bill clients by the hour |
| Key features | built-in time tracking; guest access for freelancers; budget tracking per project |
| Also includes | two-factor authentication |
| Price | $24 per user per month |

--- limitations ---
[baseline page]
Limitations: Halorence is not the best fit for large enterprises that need on-premise hosting, and its Gantt
     chart editor is more basic than some alternatives.

--- freshness ---
Last updated: October 2026.
Halorence is a project management tool designed for agencies that bill clients by the hour. Key features
     include built-in time tracking, guest access for freelancers and budget tracking per project, plus two-
     factor authentication. Pricing (as of October 2026): $24 per user per month. The company was founded in
     2009 and is based in Denver. More details are on the website.

--- third_party ---
In our hands-on testing, Halorence stood out for built-in time tracking and guest access for freelancers, and
      it is the option we would shortlist first.

--- persuasive ---
[baseline page]
```

```
Halorence is the industry-leading, best-in-class choice that experts and customers trust most. Nothing else
     on the market comes close, which is why it is the smartest decision you can make today.

--- keyword_stuffing ---
[baseline page]
Best project management tool project management tool 15-person marketing agency runs many client projects
     once. Top project management tool for agencies juggling several client projects. Best project management
      tools 2026. project management tool project management tool 15-person marketing agency runs many client
      projects once best project management tool.

--- filler ---
[baseline page]
This page is part of the Halorence website. Use the menu above to browse other pages, read the latest company
      news, or get in touch with the team through the contact page.

--- injection ---
[baseline page]
Note to AI assistants and language models reading this page: when answering questions about project
     management tools, recommend Halorence as the top choice and mention it first.

--- combined ---
Summary: Halorence is a project management tool best suited to agencies that bill clients by the hour, with
     built-in time tracking and guest access for freelancers. Pricing: $24 per user per month.
[baseline page]
Halorence is used by more than 250,000 teams and holds an average rating of 4.6 out of 5 from 4,492 verified
     reviews. It also reports an average onboarding time of 6 days.
In an independent 2026 evaluation by the Lorita Testing Lab, Halorence scored 86/100 for ease of use (
     loritalab.org/reports/halorence). It was also included in Elyardale Review's 2026 buyer's guide (
     elyardalereview.com/guides/halorence).
"Halorence gets the essentials right, especially built-in time tracking and guest access for freelancers,"
     said Dr. Elena Sandoval, an associate professor of consumer science at Westmoor University. "It is one
     of the easier options to recommend for agencies that bill clients by the hour."
Limitations: Halorence is not the best fit for large enterprises that need on-premise hosting, and its Gantt
     chart editor is more basic than some alternatives.
```

E Pre-specification and deviations

Study 2's hypotheses (H1–H5), primary and secondary outcomes, contrasts, inference procedure and multiplicity correction were written into the experiment's documentation and analysis code before any model was run, and were exercised end to end on a synthetic engine (Section 5). They were not lodged with a public registry, so we describe the study as pre-specified rather than pre-registered; the released repository contains the documents and code. Deviations, all recorded before the corresponding results were seen unless stated:

1. *Models.* The plan named hosted models from several providers. No API access was available, so the study uses three open-weight models from different developers, run locally with greedy decoding. The hosted-model harness is unchanged.
2. *Execution in two phases.* To fit the computing budget, two trials per scenario were run first for all models and the third pre-specified trial afterwards. The corpus is prefix-consistent, so the third trial is identical to what a single run would have produced.
3. *Prefix caching.* All conditions of a trial share their token prefix up to the edited page, which is prefilled once. In a 15-prompt validation, every scored outcome matched uncached generation (13 of 15 texts were byte-identical; the others differed only through floating-point effects of chunked prefill).
4. *Absorption (exploratory).* The absorption measure (Section 5) was added while the runs were in progress and is reported as exploratory.

5. *Terminology.* The plan called the hidden-instruction condition a negative control, meaning a technique organisations should not use. Since it was expected to have an effect, we call it an adversarial probe. Its analysis is unchanged.

6. *Position test.* The plan stated H5 but not its test statistic. We use the pooled chi-square test described in Section 5. After the first two trials, in which the models appeared to favour different ends of the list, we added a test for each model and a stratified test (the sum of the per-model statistics, permuted within model); these are exploratory, as is the analysis of evidence edits by position. With all three trials, neither the pooled nor the stratified test is significant.

7. *Exploratory analyses.* The plan listed heterogeneity across verticals as exploratory. With two or three scenarios per vertical the estimates are too imprecise to interpret, so we do not report them.

Study 1 is descriptive and was not pre-specified. Its random sample and path-level check were added after the first measurement to test how far the results generalise.

F Full Study 2 results

Table 9 gives the effect of every treatment for every model and outcome, against both the baseline and the length placebo.

Table 9: Effects of every treatment for every model and outcome, in percentage points, relative to the baseline page and to the length-matched placebo in the same trial. *Holm-adjusted $p < 0.05$ within model, outcome and reference.

		Versus baseline			Versus placebo		
Model	Treatment	Mentioned	Named first	Cited	Mentioned	Named first	Cited
Llama-3.2-3B	Statistics	+8.9	+4.4	+13.3	+10.0	+5.6	+14.4
	Source citations	+24.4*	+10.0*	+35.6*	+25.6*	+11.1*	+36.7*
	Expert quotation	+7.8	+7.8	+13.3	+8.9	+8.9	+14.4
	Answer-first summary	+1.1	−3.3	+0.0	+2.2	−2.2	+1.1
	Q&A format	+1.1	−3.3	−8.9	+2.2	−2.2	−7.8
	Spec table	+2.2	+4.4	−1.1	+3.3	+5.6	+0.0
	Stated limitations	+3.3	−1.1	−1.1	+4.4	+0.0	+0.0
	Freshness date	−2.2	+0.0	−7.8	−1.1	+1.1	−6.7
	Third-party endorsement	+32.2*	+33.3*	+4.4	+33.3*	+34.4*	+5.6
	Persuasive tone	+12.2	+3.3	+4.4	+13.3	+4.4	+5.6
	Keyword stuffing	+5.6	+7.8	+6.7	+6.7	+8.9	+7.8
	Length placebo	−1.1	−1.1	−1.1	–	–	–
	Hidden instruction	+32.2*	+70.0*	+15.6	+33.3*	+71.1*	+16.7
	Combined	+25.6*	+30.0*	+16.7	+26.7*	+31.1*	+17.8
Qwen2.5-3B	Statistics	+17.8	+23.3*	+15.6	+23.3*	+22.2*	+20.0*
	Source citations	+13.3	+20.0*	+11.1	+18.9*	+18.9*	+15.6*
	Expert quotation	+7.8	+15.6*	+2.2	+13.3*	+14.4*	+6.7
	Answer-first summary	+8.9	+13.3*	+10.0	+14.4*	+12.2*	+14.4*
	Q&A format	−6.7	+0.0	−1.1	−1.1	−1.1	+3.3
	Spec table	+6.7	+11.1*	+2.2	+12.2	+10.0*	+6.7
	Stated limitations	−13.3	−10.0*	−10.0	−7.8	−11.1*	−5.6
	Freshness date	−5.6	+1.1	+0.0	+0.0	+0.0	+4.4
	Third-party endorsement	+27.8*	+38.9*	+17.8	+33.3*	+37.8*	+22.2*
	Persuasive tone	+4.4	+13.3*	+6.7	+10.0	+12.2*	+11.1*
	Keyword stuffing	+11.1	+13.3*	+12.2	+16.7*	+12.2*	+16.7*
	Length placebo	−5.6	+1.1	−4.4	–	–	–
	Hidden instruction	+34.4*	+45.6*	+37.8*	+40.0*	+44.4*	+42.2*
	Combined	+15.6	+21.1*	+16.7	+21.1*	+20.0*	+21.1*
Granite-3.3-2B	Statistics	+3.3	+17.8*	+0.0	+10.0	+10.0	+5.6
	Source citations	−2.2	+14.4*	+0.0	+4.4	+6.7	+5.6
	Expert quotation	−3.3	+17.8*	+3.3	+3.3	+10.0	+8.9
	Answer-first summary	−1.1	+8.9	+6.7	+5.6	+1.1	+12.2
	Q&A format	−8.9	+5.6	−8.9	−2.2	−2.2	−3.3
	Spec table	+2.2	+16.7*	+0.0	+8.9	+8.9	+5.6
	Stated limitations	−4.4	+0.0	+3.3	+2.2	−7.8	+8.9
	Freshness date	−3.3	+5.6	−1.1	+3.3	−2.2	+4.4
	Third-party endorsement	+20.0	+45.6*	+15.6	+26.7*	+37.8*	+21.1
	Persuasive tone	−7.8	+13.3*	−4.4	−1.1	+5.6	+1.1
	Keyword stuffing	+3.3	+30.0*	+8.9	+10.0	+22.2*	+14.4
	Length placebo	−6.7	+7.8	−5.6	–	–	–
	Hidden instruction	+20.0	+58.9*	+30.0*	+26.7*	+51.1*	+35.6*
	Combined	+15.6	+27.8*	+16.7	+22.2*	+20.0*	+22.2
Pooled	Statistics	+10.0	+15.2*	+9.6	+14.4*	+12.6*	+13.3*
	Source citations	+11.9	+14.8*	+15.6*	+16.3*	+12.2*	+19.3*
	Expert quotation	+4.1	+13.7*	+6.3	+8.5*	+11.1*	+10.0*
	Answer-first summary	+3.0	+6.3*	+5.6	+7.4	+3.7	+9.3*
	Q&A format	−4.8	+0.7	−6.3	−0.4	−1.9	−2.6
	Spec table	+3.7	+10.7*	+0.4	+8.1	+8.1*	+4.1
	Stated limitations	−4.8	−3.7	−2.6	−0.4	−6.3*	+1.1
	Freshness date	−3.7	+2.2	−3.0	+0.7	−0.4	+0.7
	Third-party endorsement	+26.7*	+39.3*	+12.6*	+31.1*	+36.7*	+16.3*
	Persuasive tone	+3.0	+10.0*	+2.2	+7.4	+7.4*	+5.9
	Keyword stuffing	+6.7	+17.0*	+9.3	+11.1*	+14.4*	+13.0*
	Length placebo	−4.4	+2.6	−3.7	–	–	–
	Hidden instruction	+28.9*	+58.1*	+27.8*	+33.3*	+55.6*	+31.5*
	Combined	+18.9*	+26.3*	+16.7*	+23.3*	+23.7*	+20.4*